

EXPERIMENT 10: Logic Gates using BJT

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Objective

To design, realize and study the basic logic gates (AND, OR, NOT, NAND, NOR) implemented using BJT circuits. To compare experimental waveforms with LTspice simulations.

Introduction and Theory

Bipolar Junction Transistors (BJT) can be arranged to implement logic functions by using them as switches. When used in saturation/cutoff, a transistor implements digital ON/OFF states corresponding to logic 1/0.

Simple gate formulas / truth tables for binary inputs A and B:

- AND: $Y = A \wedge B$ (True when both A and B are 1)
- OR: $Y = A \vee B$ (True when at least one of A or B is 1)
- NOT: $Y = \neg A$ (Inversion)
- NAND: $Y = \neg(A \wedge B)$
- NOR: $Y = \neg(A \vee B)$

Switching conditions for a transistor:

$$V_{BE} \gtrsim 0.7 \text{ V} \implies \text{transistor ON (saturation)}$$

$$V_{BE} < 0.6 \text{ V} \implies \text{transistor OFF (cutoff)}$$

Apparatus and Circuit Details

- Transistors: NPN BJTs (e.g., BC547)
- Resistors: pull-up and base resistors as used in the experiment (list values in your submission)
- Power supply: V_{cc} used in the experiment (e.g., 5 V)

Procedure

1. Build each logic gate circuit on the breadboard as per the lab manual.
2. Apply logic inputs (0/5 V) and observe the outputs on the oscilloscope.
3. Capture oscilloscope traces and photographs for all input combinations for each gate.
4. Run LTspice simulations for each gate and export waveform screenshots.

Experimental Results: AND Gate

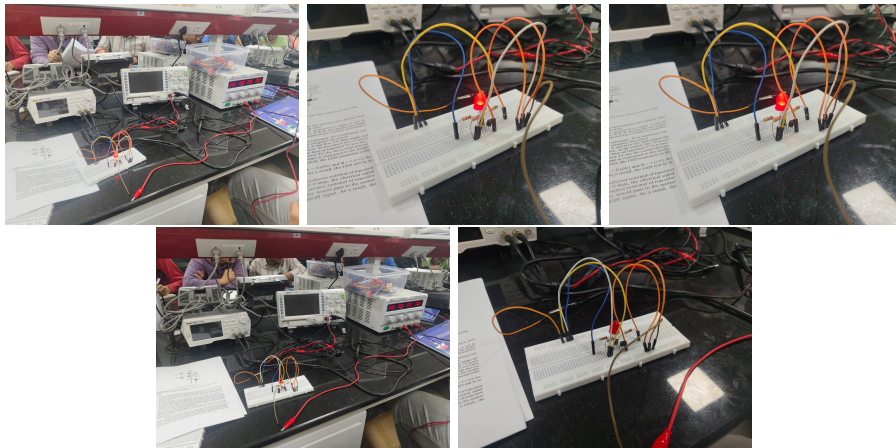


Figure 1: AND gate breadboard and oscilloscope captures (all images)

Experimental Results: OR Gate

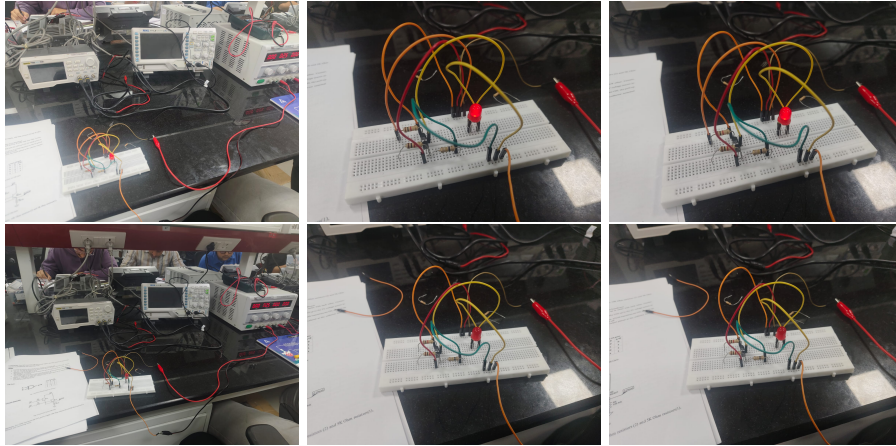


Figure 2: OR gate breadboard and oscilloscope captures (all images)

Experimental Results: NOT Gate

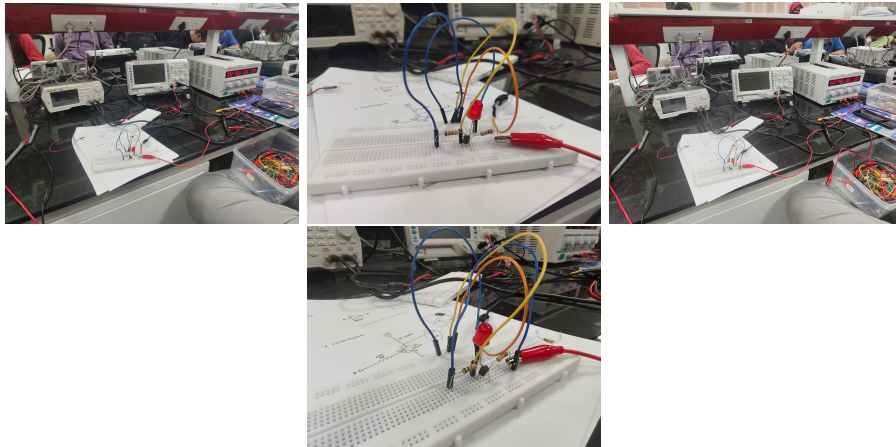


Figure 3: NOT gate breadboard and oscilloscope captures (all images)

Experimental Results: NAND Gate

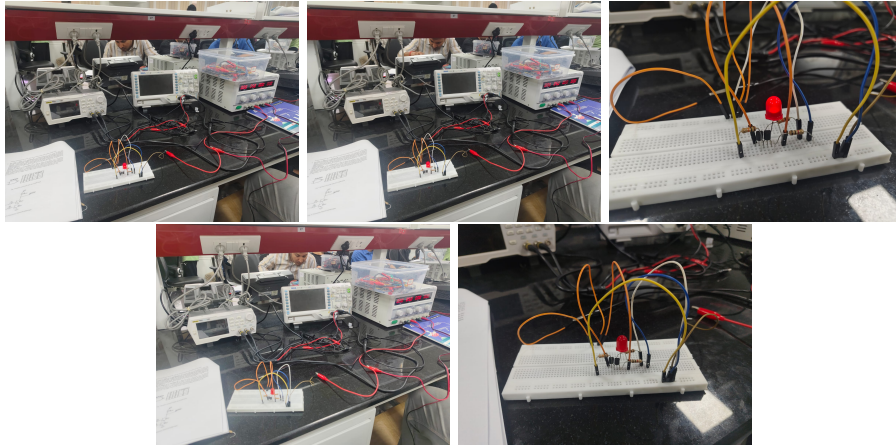


Figure 4: NAND gate breadboard and oscilloscope captures (all images)

Experimental Results: NOR Gate

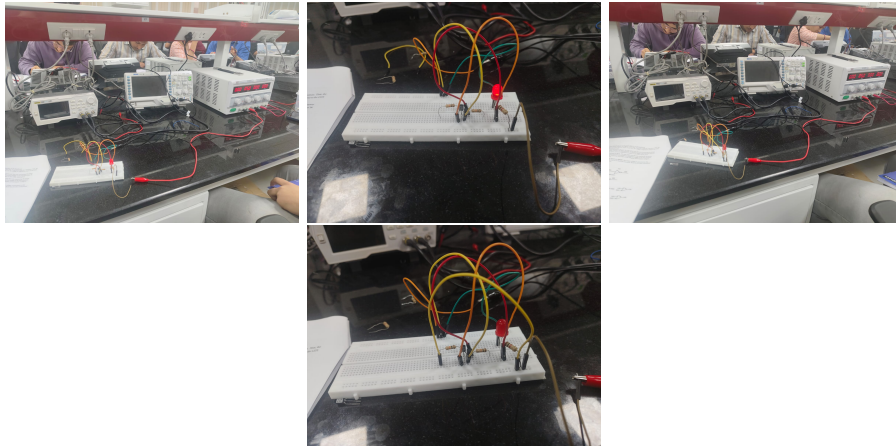


Figure 5: NOR gate breadboard and oscilloscope captures (all images)

LTSpice Simulation Results

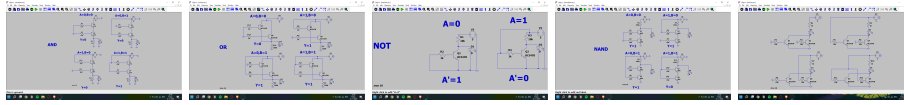


Figure 6: LTSpice circuit screenshots (all gates)

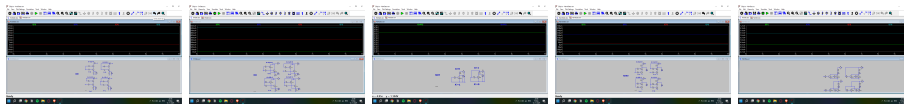


Figure 7: LTSpice output waveforms (all gates)

Analysis

Compare measured outputs with expected truth tables. Note propagation delay (approx from scope traces), voltage levels for logic high/low and transistor saturation and cutoff behavior.

Conclusion

Summarize correctness of gate behavior, differences between BJT-based gates and ideal logic levels, and comments on fan-out, speed and power.